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Article in *AIDS Research and Human Retroviruses* · October 2021

DOI: 10.1089/AID.2021.0157

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HIV-1 Drug Resistance Among Treatment-Naive Transgenders from India

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Abstract

Transgenders (TGs) are highly affected by HIV with high prevalence of 3.14% in India. Since 2017, targeted preventive efforts have been initiated by the government and HIV-infected TGs are being provided the anti-retroviral (ART) treatment. Information on the primary HIV drug resistance is crucial for appropriate treatment selection to curb further spread of HIV in this population. In this study, we analyzed HIV-1 *pol* gene sequences from 36 TGs for presence of drug resistance mutations. To our knowledge, this first study from India reports high-level primary drug resistance (13.8%) among the TG population. Mutations M184V, A98G, K103N, G190A, and Y318F associated with resistance to nucleoside reverse transcriptase inhibitor and non-nucleoside reverse transcriptase inhibitors were observed. All *pol* gene sequences revealed HIV-1 subtype C in all study TG. High-level HIV-1 drug resistance warrant nationwide larger studies on TGs to understand the level of primary ART drug resistance among this population.

Keywords: transgender, HIV-1, primary drug resistance, India

GLOBAL HIV PREVALENCE among the transgender (TG) people is reported to be varied ranging from 8% to 68%¹ counting nearly 25 million.² In India, the TG people are highly affected by HIV. The HIV prevalence in TG is much higher (3.14%) than the general population as well as men who have sex with men (2.69%) and female sex workers (1.56%).³ TG people experience high-level stigma and discrimination, violence, and abuse, and are inclined to sex work for income as they face economic challenges. Around 90% of TG in India are involved in sex work making them more vulnerable to HIV infection.⁴ There has been improved access to HIV services for TG population after the decision of Indian Supreme Court to recognize them as a distinct gender.

The targeted preventive efforts have been initiated by National AIDS Control Organization and by 2017, around 6792 TG were registered with free antiretroviral (ART) program.⁵

Although TG people in India are being linked to treatment and care, data on the levels of primary HIV drug resistance as well as HIV genotypes in this population are lacking.

In this study, we analyzed the primary drug resistance among TG ($N=50$) who were diagnosed with HIV infection and initiating the antiretroviral treatment for the first time. The HIV-1 genotypes were assessed through phylogenetic analysis using the HIV-1 *pol* gene sequences. To our knowledge, this is the first report on primary HIV-1 drug resistance and HIV-1 genotypes among Indian TGs.

The study was approved by the institutional Ethics Committee (Protocol Number: NARI-EC/2017-15). Dried blood spot (DBS) specimens were collected from five HIV care centers that were following up quarterly with the TG for HIV testing. Written informed consent was obtained from all study participants before sample collection. The TG

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diagnosed with HIV-1 infection for the first time ($N=50$, median age: 25 years; range: 20–45 years) were enrolled in the study. Demographic information of the study participants is mentioned in Table 1.

The DBS specimens were processed for extraction of total nucleic acids and further amplification and sequencing of partial *pol* gene (comprising complete protease and partial reverse transcriptase genes) using the in-house genotyping protocol validated by WHO for DBS based genotyping. In brief, nucleic acids were extracted from two spots of the DBS specimens using the automated NucliSens Easy Meg nucleic acid extractor (Biomeriux, Germany). The nucleic acids were then subjected to nested reverse transcriptase polymerase chain reaction using primers spanning HXB2 genome 2135 to 3583 comprising complete protease and partial reverse transcriptase gene (1-upto 343 amino acids). The amplified products were then purified and sequenced using Genetic Analyzer (Applied Biosystems 3130XL; Thermo Fisher Scientific) using the same set of outer primers. Sequences were aligned and analyzed using SeqScape v2.7 (Thermo Fisher Scientific). The drug resistance interpretation was performed using the Stanford HIV drug resistance database (hivdb.stanford.edu; algorithm version 8.6.1 dated July 18, 2018). Presence of major, minor, or accessory mutation was considered as resistance mutation.

The HIV genotypes were assessed by phylogenetic analysis using the *pol* gene sequences. The phylogenetic analysis of *pol* gene sequences was carried out with HIV-1 reference sequences of subtype C and other subtypes that were retrieved from Los Alamos HIV sequence database (<https://www.hiv.lanl.gov/content/sequence/NEWALIGN/align.html>). The phylogenetic tree was constructed using the software MEGA 6.0.⁶ Transmission cluster analysis was performed by Cluster picker v 1.2.5 where the clusters were designated on the basis of sequences having genetic distance of $\leq 4.5\%$ with bootstrap support of $\geq 90\%$ for grouping of ≥ 2 sequences.

In this preliminary study, 36/50 specimens could be successfully amplified for the *pol* gene. The HIV drug resistance interpretation showed presence of mutations associated with ART drug resistance in 5/36 (13.8%) specimens. Resistance to reverse transcriptase inhibitors was seen in 3/33 (8.3%) samples, whereas 2 samples (5.5%) showed presence of ac-

cessory mutation in the protease gene. Major mutation associated with resistance to nucleoside reverse transcriptase inhibitors (NRTIs) observed was M184V, whereas Thymidine Analog Mutation (TAM) D67N was seen in one sample. The frequency of mutations is shown in Figure 1a. Resistance to non-nucleoside reverse transcriptase inhibitors (NNRTIs) was shown by mutations A98G, K103N, Y318F, and G190A. Of the five samples, one sample showed presence of mutations M184V, A98G, K103N, and Y318F conferring resistance to NRTI and NNRTIs. Mutation G190A was seen in one sample, whereas two samples had mutations K20KT and D67DN associated with PR and NRTI resistance, respectively. The phylogenetic analysis of *pol* gene sequences of the 36 samples revealed HIV-1 subtype C as the genotype (Fig. 1b). All sequences clustered together with Indian HIV-1 subtype C reference sequence. No recombinant was observed.

The increasing HIV prevalence among the TGs in India is alarming. India is one of the countries that have targeted TG people and successfully providing HIV services for control of further spread. With the targeted intervention approach, TGs are now linked to HIV testing services. By March 2019, 92% of the identified TG population has been initiated on ART.⁷ However, information on primary drug resistance is one of the crucial factors to understand the cause of the high prevalence. In this exploratory study, we report high-level drug resistance (13.8%) associated with first-line ART drugs given in India under the free ART program making the treatment ineffective. The mutations associated with NNRTI and NRTI is indicative of resistance to Nevirapine and Lamivudine that are included in the first-line regimen. Mutation M184V is known to confer high-level resistance to Lamivudine, whereas A98G, K103N, G190A, and Y318F reduce the susceptibility for Nevirapine as well as Efavirenz.^{8–11} The accessory mutations K20T and I47M are known to be associated with reduced susceptibility to protease inhibitors.¹² Mutation I47M is a rarely observed atypical mutation reported in a study from Iran.¹³ These findings highlight presence of high-level drug resistance to the ART drugs given as first-line regimen in India. ART treatment outcomes vary as per the HIV subtypes.¹⁴ Information on HIV-1 genotypes in Indian TG population is lacking. Our study reports presence of HIV-1 subtype C in all 36 samples, which is predominant in India.¹⁵ Although these sequences align with subtype C reference sequences, all 36 sequences cluster together with the Indian subtype C reference sequence. Subtype analysis based on multiple genes would be essential to identify any recombinants in this population. Considering the fact that the TG population is highly mobile and involved in sex work with multiple sexual partners, wider studies spanning nationwide TG population with critical full genome sequence analysis are needed. The transmission cluster analysis revealed five clusters of two sequences (Fig. 1c, d) where the intracluster genetic distance for these clusters ranged between 2.9% and 4.3%. Cluster 8 had two sequences from two different places, whereas sequences in other clusters were from the same place. TG people have very compact and closed communities. Hence, intracluster transmission analysis would help identifying the HIV circulation among them.

This exploratory study reports for the first time the primary drug resistance and HIV genotypes in Indian TG population. Small sample size is a limitation of our study and studies on

TABLE 1. DEMOGRAPHIC INFORMATION OF TRANSGENDER STUDY PARTICIPANTS

Description	No. of TGs
Age	20–30 years: 28 Median age: 25 years
	31–40 years: 6
	41–45 years: 2
Education	Noneducated: 10 Higher secondary: 23 Graduation: 3
Monthly income (INR)	0–10,000: 19 11,000–25,000: 15 No information: 2
Frequency of sexual intercourse (per week)	<50: 31 >50: 5

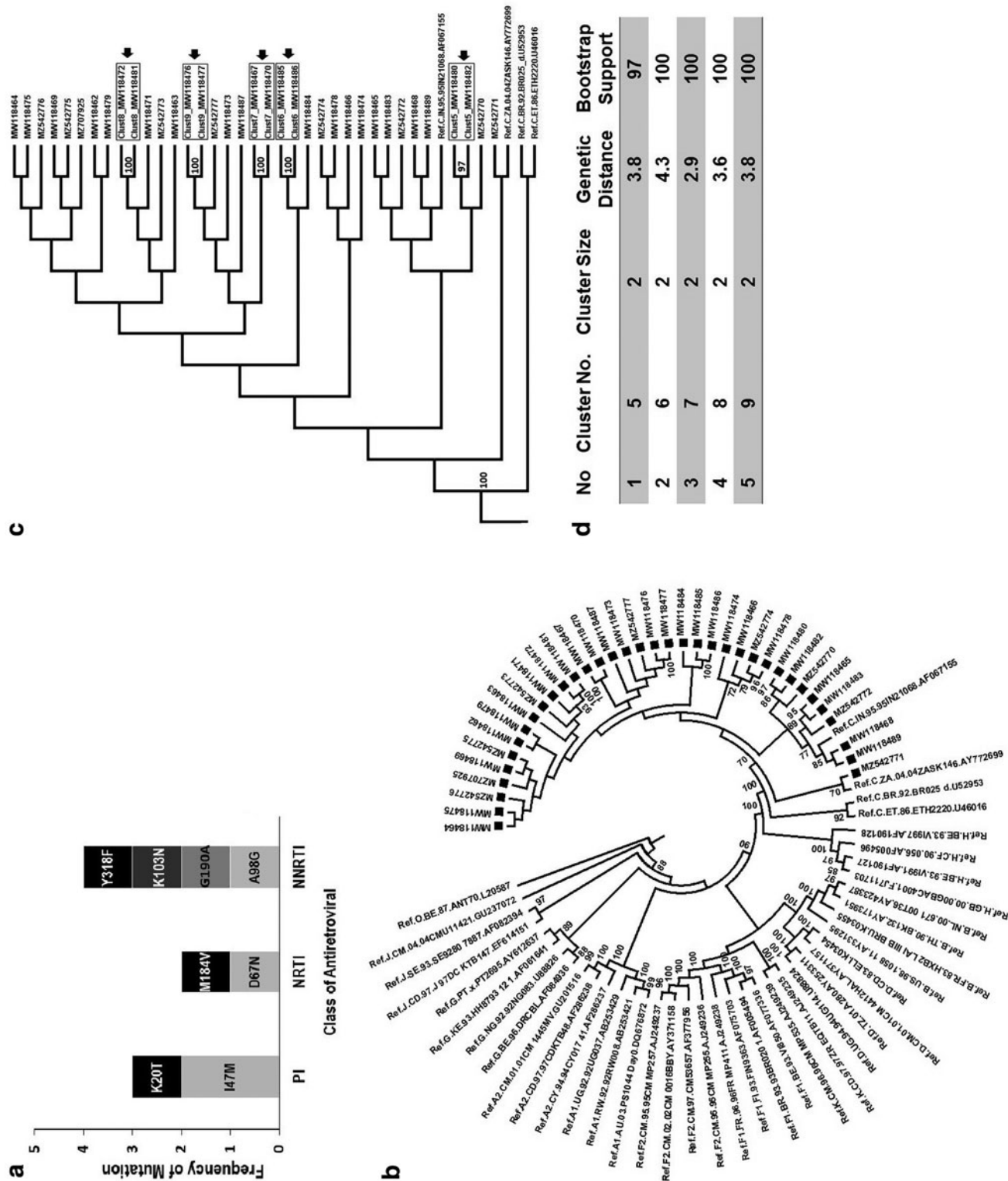


FIG. 1. Frequency of drug resistance mutations and phylogenetic analysis. **(a)** Frequency of mutations associated with drug resistance. **(b)** Phylogenetic analysis of HIV-1 pol gene sequences from transgender ($N = 36$). The study sequences are marked with ■. **(c)** Transmission cluster analysis identified five clusters of the sequences with genetic distances across each cluster **(d)**.

larger number of TG population across India are essential for correct interpretations and appropriate treatment selection for this key population.

Sequence Data

The GenBank accession numbers for 36 pol gene sequences are MW118462–MW118487, MW118487, MZ542770–MZ542777, and MZ707925.

Authors' Contributions

S.N.K. conceived and designed the study, and drafted the article. S.K., S.R., S.M., A.V., and U.C. coordinated and collected the samples. S.G. performed the experiments. S.N.K., S.G., and A.P. analyzed the data. U.C., V.A., and R.G. critically revised the article. All authors read and approved the final article.

Acknowledgments

The authors thank all the study participants and the staff at the centers for their support in enrollment of participants and sample collection.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

This study was done through the institutional intramural grant of ICMR-National AIDS Research Institute, Pune.

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